AutoApply
PRO PLATFORM

AutoApply Pro

AI-Powered Career & Recruitment Platform

Full-Stack Development Plan · Multi-Role Portals · Email Automation

Infrastructure Budget · Monthly Operational Costs

Phase 1–4 Roadmap | 4 User Roles | Email Engine | Budget Analysis

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1. Executive Summary

AutoApply Pro is an AI-powered career and recruitment SaaS platform designed to eliminate the friction in job searching and hiring. At its core, it automates the most tedious parts of the recruitment lifecycle — from extracting recruiter contact details out of PDFs or web pages, to drafting and sending personalised outreach emails at scale, to tracking every application through its entire lifecycle.

Mission: Reduce a candidate's average job-search cycle from 6 months to under 6 weeks by combining smart email automation, AI-driven matching, ATS optimisation, and real-time market intelligence — all in a single, role-aware platform.

Metric	Target (Year 1)	Target (Year 2)
Registered Candidates	5,000	25,000
Active Recruiters / Companies	200	1,500
Emails Sent / Month	50,000	500,000
Avg. Time-to-Interview	3 weeks	10 days
Platform Uptime SLA	99.5%	99.9%

2. Platform Overview & Vision

AutoApply Pro is structured as a **multi-sided marketplace** — candidates get automation tools; recruiters get pre-screened talent pipelines; companies get employer branding and analytics; admins get full governance control. All four portals share a single backend API and data layer, ensuring consistency while serving radically different user needs.

Portal	Primary Users	Core Value
■ Candidate	Job seekers, fresh graduates, experienced professionals	AI-driven auto-apply, resume optimisation, interview prep
■ Recruiter	Independent headhunters, staffing agencies	Candidate discovery, pipeline management, bulk outreach
■ Company	HR teams, hiring managers, employer brand leads	Job posting, ATS integration, talent analytics
■ Admin	Platform operators, compliance officers	User governance, content moderation, billing oversight

3. Email Extraction & Automated Outreach Engine

Core Requirement: Users can upload a PDF, paste a company URL, or enter a LinkedIn/Indeed page. The platform automatically extracts recruiter/hiring-manager email addresses, generates personalised outreach emails using AI, and sends them from the user's own device (SMTP) or via the platform's managed email service.

3.1 Email Extraction Methods

■ PDF Upload Extraction

User uploads a resume, company brochure, or job-posting PDF. The backend uses pdfplumber / PyMuPDF to parse text and apply regex patterns (plus an ML NER model for context-aware extraction) to identify email addresses. Extracted emails are deduplicated and shown in a review panel before sending.

■ URL / Web Scraping

User inputs a company website or LinkedIn company page URL. A headless Playwright/Puppeteer scraper visits the page, extracts visible emails, and also infers likely patterns (e.g. firstname.lastname@company.com) from domain + name data. Respects robots.txt and rate limits.

■ CSV / Contact List Import

Bulk import of recruiter contact lists via CSV. The system validates, deduplicates, and enriches records using Hunter.io / Apollo.io APIs.

■ Manual Entry

Direct entry for known contacts. AI suggests additional contacts from the same company domain using pattern inference.

3.2 Email Generation & Sending Pipeline

1 – Extract	Parse PDF / URL → find emails	pdfplumber, Playwright, Regex + NER
2 – Enrich	Verify email validity, add name/role	Hunter.io API, NeverBounce
3 – Draft	AI generates personalised email body	Claude / GPT-4o API
4 – Review	User previews & edits draft	React rich-text editor
5 – Schedule	Set send time, daily limits	Bull/Redis queue
6 – Send	SMTP relay or managed transactional	User's Gmail/Outlook or SendGrid
7 – Track	Open/click/reply detection	Pixel tracking + webhook
8 – Log	Update application status in DB	PostgreSQL + Prisma ORM

■ ***Deliverability safeguards:*** *Rate-limiting (max 50 emails/hr per user account), domain warm-up schedule, SPF/DKIM/DMARC configuration guidance, unsubscribe link injection, and bounce/spam-complaint suppression lists are all*

enforced by the platform to protect sender reputation.

4. Multi-Role Portal Architecture

Each portal shares the same React frontend codebase but uses **role-based rendering** — a single /dashboard route renders completely different components depending on the authenticated user's role (CANDIDATE | RECRUITER | COMPANY | ADMIN). Backend APIs enforce role-level permissions via JWT claims.

4.1 Candidate Portal

The candidate portal is the flagship experience — designed for both fresh graduates and experienced professionals who want to supercharge their job search.

- **Profile Builder:** Multi-profile support — e.g. separate profiles for 'Java Developer' vs 'Python ML Engineer' with distinct resumes, skill tags, and cover letter templates.
- **Resume Builder & ATS Optimizer:** Drag-and-drop resume builder with real-time ATS score feedback, keyword gap analysis vs job description, and one-click tailoring.
- **Auto-Apply Engine:** Upload a PDF or paste a URL → extract recruiter emails → AI drafts personalised cold emails → user reviews → scheduled send from their own email account.
- **Job Discovery Feed:** Aggregated feed from LinkedIn, Indeed, Naukri (India), and Internshala with AI-powered match scores based on skills and preferences.
- **Application Tracker:** Kanban board view: Wishlist → Applied → Shortlisted → Interview → Offer → Rejected. Automatic status updates via email webhook parsing.
- **Interview Prep Hub:** Company-specific research cards, common interview questions bank, AI mock interview with voice + text support, performance scoring.
- **Market Insights:** Trending skills for your role, salary benchmarks, city-wise demand heatmaps, and upskilling roadmap recommendations.

4.2 Recruiter Portal

Built for independent recruiters and staffing agencies who manage multiple client requisitions simultaneously.

- **Candidate Search & Filter:** Boolean search across the candidate database by skills, experience, location, notice period, and AI-inferred culture-fit score.
- **Saved Searches & Alerts:** Save search criteria and receive daily digest emails when new matching candidates join the platform.
- **Bulk Outreach:** Select up to 500 candidates → AI personalises each outreach message with the candidate's name and key matching skill → send via platform SMTP relay.
- **Pipeline Manager:** Drag-and-drop talent pipeline per job requisition with stage-wise notes, interview scheduling integration, and client-sharing links.
- **Candidate Profile View:** Rich candidate cards with parsed resume, LinkedIn-style portfolio, GitHub activity heatmap, and AI-generated candidate summary.
- **Commission Tracker:** Log placements and track commission milestones per requisition with invoice generation support.

4.3 Company Portal

Designed for HR teams and hiring managers at small-to-enterprise companies wanting to manage their talent acquisition in one place.

- **Employer Branding Page:** GitHub-style public company page with culture videos, team photos, employee testimonials, tech stack showcase, and perks listing.
- **Job Posting & Distribution:** Create job descriptions with AI assist → auto-distribute to integrated job boards (LinkedIn, Indeed, Naukri) with one click.
- **Applicant Tracking System (ATS):** Full-featured ATS with configurable hiring stages, structured interview scorecards, offer letter generation, and onboarding checklist.
- **Analytics Dashboard:** Time-to-hire metrics, source-of-hire breakdown, offer acceptance rate, diversity hiring analytics, and recruiter performance scorecards.
- **Team Collaboration:** Multiple HR team members with role-based access (HR Admin, Hiring Manager, Interviewer) and shared candidate notes.
- **Bulk Email to Applicants:** Automated rejection/shortlist/interview invite emails with customisable templates and personalisation tokens.

4.4 Admin Portal

Full platform governance — used by the AutoApply Pro operations team.

- **User Management:** View, search, suspend, ban, or verify any user account across all roles. Bulk actions, audit logs, and impersonation mode for support.
- **Content Moderation:** Flag/review/approve job postings, company pages, and candidate profiles for spam, inappropriate content, or policy violations.
- **Email Compliance Monitor:** Monitor aggregate email sending volumes, bounce rates, and spam complaint rates per user. Auto-suspend accounts that breach thresholds.
- **Billing & Subscription Management:** View all subscriptions, process refunds, apply discount codes, and generate revenue reports. Integrated with Stripe.
- **Platform Analytics:** DAU/MAU, feature adoption funnels, email send volumes, API usage, infrastructure cost per user, and revenue attribution.
- **System Health Dashboard:** Real-time monitoring of API latency, database performance, email queue depth, error rates, and uptime metrics.

5. Development Roadmap (Phase 1–4)

Phase 1	Foundation — Email Outreach Engine	Months 1–3

Goal: Launch the core email automation MVP with basic job tracking.

✓ User authentication (OAuth + email/password) for all 4 role types
✓ Candidate profile builder with multi-profile support
✓ PDF upload → email extraction engine (pdfplumber + regex + NER)
✓ URL scraping for recruiter email extraction (Playwright)
✓ AI email drafting using Claude/GPT-4o API
✓ SMTP-based sending from user's Gmail / Outlook (OAuth2 secure connect)
✓ Platform-managed transactional sending via SendGrid
✓ Rate limiting, scheduling queue (Bull + Redis)
✓ Basic application tracker (Kanban: Sent / Pending / Replied)
✓ PostgreSQL database schema + Prisma ORM setup
✓ Admin panel: user management + email compliance monitor

Phase 2	Discovery & Intelligent Matching	Months 4–6

Goal: Automate job discovery and introduce AI-powered matching.

✓ Job aggregator agents: LinkedIn Jobs API, Indeed Publisher, Naukri scraper
✓ AI match scoring engine: LLM-based JD vs candidate skills analysis
✓ Personalised job feed for candidates with match percentage
✓ Recruiter portal: candidate search, boolean filters, pipeline manager
✓ Enhanced tracker: ATS score analysis, rejection reason classification
✓ Company portal: job posting → multi-board distribution
✓ Email open/click/reply tracking via pixel + webhook
✓ Bulk outreach for recruiters (up to 500 contacts)
✓ Email verification integration (NeverBounce / ZeroBounce)
✓ Mobile-responsive frontend (PWA)

Phase 3	Portfolio & Resume Optimisation	Months 7–9

Goal: Comprehensive profile management and resume intelligence.

✓ Drag-and-drop resume builder with 10+ ATS-friendly templates
✓ Real-time ATS score checker with keyword gap analysis
✓ GitHub-style public candidate portfolio page
✓ Company employer branding page with media uploads
✓ Full ATS for company portal: structured scorecards, offer letters
✓ Interview prep hub: company research cards, Q&A; bank

✓ AI mock interview (text + optional voice via browser Web Speech API)
✓ Referral programme module
✓ Stripe billing integration for premium plans
✓ Company analytics dashboard v1

Phase 4	Advanced Career Coaching & Market Intelligence	Months 10–12

Goal: Full-lifecycle career support with data-driven insights.

✓ Market insight dashboard: trending skills, salary benchmarks, demand heatmaps
✓ AI upskilling roadmap with curated course recommendations
✓ Diversity & inclusion analytics for companies
✓ Advanced email personalisation using candidate activity signals
✓ Integration with Calendly / Google Calendar for interview scheduling
✓ Commission tracker + invoice generation for recruiters
✓ White-label option for enterprise clients
✓ Public API + webhook system for third-party integrations
✓ SOC 2 Type I audit preparation
✓ Native iOS & Android app (React Native)

6. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Next.js 14, TypeScript, TailwindCSS	SSR/SSG, responsive UI, SEO
State Mgmt	Zustand + React Query (TanStack)	Client state + server data caching
Backend API	Node.js + Express / Fastify (TypeScript)	REST API + GraphQL (Phase 3+)
Auth	NextAuth.js + JWT + OAuth2 (Google, LinkedIn)	Secure multi-role auth
Database	PostgreSQL 15 (primary) + Redis (cache/queue)	Relational data + email queues
ORM	Prisma	Type-safe DB queries
Email Sending	Nodemailer (SMTP) + SendGrid API	User SMTP relay + managed delivery
Email Parsing	pdfplumber, PyMuPDF (Python microservice)	PDF email extraction
Web Scraping	Playwright (Node.js)	URL-based email extraction
AI / LLM	Anthropic Claude API / OpenAI GPT-4o	Email drafting, matching, coaching
Job Boards	LinkedIn Jobs API, Indeed Publisher, Naukri RSS	Job discovery aggregation
Email Verify	NeverBounce / Hunter.io APIs	Email validation & enrichment
File Storage	AWS S3 / Cloudflare R2	Resumes, company media
Queue	BullMQ + Redis	Email scheduling, background jobs
Monitoring	Sentry + Datadog / Grafana + Prometheus	Error tracking, metrics
CI/CD	GitHub Actions → Vercel (frontend) + Railway/Render (backend)	Automated deploys
Payments	Stripe	Subscription billing, invoicing
Hosting	Vercel (frontend) + Railway or AWS ECS (backend)	Scalable cloud infra

7. Security & Compliance

- **OAuth2 + App Passwords:** Never store plain-text email passwords. Use Google/Microsoft OAuth2 tokens or App Passwords with encrypted storage (AES-256 at rest).
- **JWT with Short Expiry:** Access tokens: 15-min TTL. Refresh tokens: 7-day TTL, stored in httpOnly cookies. Role claims embedded in token payload.
- **Rate Limiting & Anti-Spam:** Max 50 outbound emails/hr per user, 200/day hard cap. Automatic suspension on >2% bounce rate or >0.1% spam complaint rate.
- **Data Encryption:** All data encrypted in transit (TLS 1.3) and at rest. PII fields hashed where appropriate.
- **GDPR / IT Act Compliance:** Explicit consent flows for email marketing, right-to-delete implementation, data residency options (India: Mumbai AWS region).
- **Email Unsubscribe:** Every outbound email includes a one-click unsubscribe link. Suppression list maintained globally across the platform.
- **Audit Logging:** Every sensitive action (login, email send, data export) is logged with timestamp, IP, user ID for compliance and forensics.

8. Infrastructure & Monthly Budget

All costs below are in **Indian Rupees (INR)** and **USD** where the service is priced in USD (converted at ■83/USD). Costs shown are for the **Growth Stage (post-launch, ~500 active users)**. Startup stage costs are ~40% lower.

8.1 Hosting & Infrastructure

Service	Plan / Tier	USD/mo	INR/mo	Notes
Vercel (Frontend)	Pro Plan	$20	■1,660	Next.js hosting, CDN, analytics
Railway / Render (Backend)	Team Plan	$20	■1,660	Node.js API server, auto-scale
PostgreSQL (Railway/Supabase)	Pro	$25	■2,075	Managed Postgres, daily backups
Redis (Upstash / Railway)	Pay-as-you-go	$10	■830	Queue + cache
AWS S3 / Cloudflare R2	Pay-as-you-go	$5	■415	Resume & media storage (50 GB est.)
Domain (.com)	Annual ÷12	$1.5	■124	GoDaddy / Namecheap
SSL Certificate	Free (Let's Encrypt)	$0	■0	Auto-renewed
CDN (Cloudflare Free)	Free	$0	■0	DDoS protection included
Subtotal		**$81.5**	**■6,764**	

8.2 Third-Party API & Services

Service	Usage	Free Tier	Paid Est. USD/mo	INR/mo
Claude / OpenAI API	Email drafting, matching, coaching (~50K tokens/day)	Limited	$30–60	■2,490–4,980
SendGrid (Email Relay)	Platform-managed sending, transactional	100 emails/day	$20 (Essentials)	■1,660
Hunter.io	Email finding & verification	25 searches/mo	$49 (Starter)	■4,067
NeverBounce / ZeroBounce	Email list validation	250 credits free	$16 / 5K verifications	■1,328
LinkedIn Jobs API	Job aggregation	No free tier	$0 (Partner prog.)	■0 (apply for access)
Stripe	Subscription billing	Free (2.9%+30¢ per txn)	$15 (at 100 subs	■1,245
Sentry	Error monitoring	5K events/mo	$0 initially	■0
Datadog / Grafana Cloud	Infra monitoring	Limited free	$0–$20	■0–1,660
Subtotal (mid est.)			**~$160**	**~■13,280**

8.3 Team & Operations (Optional — Bootstrap Phase)

Role	Type	INR/mo	Notes
Full-Stack Developer (you)	Co-founder / Sweat equity	—	Primary builder
UI/UX Freelancer	Contract (part-time)	■15,000–25,000	Figma designs, Phase 1–2
DevOps / Cloud setup	One-time freelancer	■10,000 one-time	CI/CD, infra setup
Content / SEO Writer	Freelance (Phase 3+)	■8,000–15,000	Blog, job descriptions
Customer Support	Self initially → hire at 500+ users		
Subtotal (lean)		**■23,000–40,000**	**Scale as revenue grows**

8.4 Monthly Expenditure Summary

Category	Startup Stage (0–200 users)	Growth Stage (~500 users)	Scale Stage (~2K+ users)
Hosting & Infra	■3,500	■6,764	■18,000+
APIs & Services	■5,000	■13,280	■35,000+
Team / Freelancers	■0–10,000	■23,000	■80,000+
Misc (domains, tools, backups)	■1,000	■2,500	■5,000
TOTAL / Month	**■9,500–14,500**	**■45,544**	**■1,38,000+**

9. One-Time Setup Costs

Item	Cost (INR)	Notes
Domain Registration (.com, 1 yr)	■800–1,200	Namecheap / GoDaddy
Logo & Brand Design	■5,000–15,000	Freelancer or Fiverr
UI/UX Design (Figma wireframes, all 4 portals)	■20,000–40,000	Freelancer
Legal: Company Registration (Pvt. Ltd.)	■10,000–15,000	LegalWiz / Vakilsearch
Privacy Policy & T&C drafting	■5,000–8,000	Legal consultant
Initial DevOps Setup (CI/CD, infra-as-code)	■10,000	One-time freelancer
Email Warm-up Tool (Lemwarm / Mailwarm)	■3,500 / 3 months	Improve deliverability
Google Workspace (Business email)	■1,400/yr	admin@autoapplypro.com etc.
Miscellaneous (testing tools, stock photos)	■5,000	
TOTAL One-Time	**■60,700–98,100**	**Budget ~■80,000–1,00,000**

10. Revenue Model & Break-Even Analysis

Subscription Tiers — Candidates

Free	■0	5 email extractions/mo, basic tracker, 1 profile, 10 AI drafts
Pro	■299/mo	Unlimited extractions, 3 profiles, 200 AI emails, ATS checker, interview prep
Elite	■599/mo	Everything in Pro + priority AI, market insights, dedicated SMTP

Subscription Tiers — Recruiters & Companies

Plan	Target	Price/mo	Key Features
Recruiter Starter	Independent recruiters	■999/mo	500 candidate views, 3 pipelines, bulk outreach 100/day
Recruiter Pro	Agencies	■2,499/mo	Unlimited views, 20 pipelines, 500 outreach/day, commission tracke
Company Basic	Startups	■1,499/mo	3 active job posts, ATS, employer page, 5 team seats
Company Growth	SMEs	■4,999/mo	Unlimited posts, full ATS, analytics, 20 seats, API access
Company Enterprise	Large corps	Custom	White-label, SLA, dedicated support, SSO, custom integrations

Break-Even Scenario (Growth Stage — Month 6)

Source	Subscribers	Avg. ARPU	Monthly Revenue
Candidate Pro	80	■299	■23,920
Candidate Elite	20	■599	■11,980
Recruiter Starter	10	■999	■9,990
Company Basic	5	■1,499	■7,495
Total Revenue			**■53,385/mo**
Total Costs (Growth)			**■45,544/mo**
Net Profit			**■7,841/mo ✓**

11. Risk Register

Risk	Likelihood	Impact	Mitigation
Emails land in spam	High	High	SPF/DKIM/DMARC setup, warm-up, rate limiting, NeverBounce validation
LinkedIn blocks scraping	High	Medium	Use official APIs where available; fall back to other job boards
AI API costs spike at scale	Medium	Medium	Implement response caching, token budgets, and tiered AI access by plan
Email provider ToS violation	Medium	High	Platform relay via SendGrid for high-volume; personal SMTP for low volume
Low conversion free→paid	Medium	High	Strong free tier; in-app upgrade prompts; annual discount offers
Data breach / PII leak	Low	Very High	Encryption at rest+transit, regular pen tests, SOC 2 prep, audit logs
Competitor replication	Medium	Medium	Network effects, Indian-market focus (Naukri, Internshala), brand trust

12. Next Immediate Actions

Focus exclusively on Phase 1 — the email extraction and outreach engine — before building anything else. Validate with real users, then iterate.

Week 1–2

- Register domain (autoapplypro.com or .in) and set up Google Workspace
- Set up GitHub monorepo with Next.js frontend + Node.js backend
- Design basic wireframes for Candidate portal (Figma or pen-paper)
- Configure PostgreSQL on Railway / Supabase free tier

Week 3–4

- Implement authentication (NextAuth + Google OAuth) for all 4 roles
- Build PDF upload API with pdfplumber email extraction (Python microservice)
- Integrate Claude API for AI email drafting
- Build basic email send via Nodemailer (user's Gmail SMTP)

Month 2

- Build Candidate dashboard: profile, application tracker (Kanban)
- Add URL scraping endpoint (Playwright) for email extraction
- Implement email scheduling queue (BullMQ + Upstash Redis)
- Soft-launch to 10 beta users — gather feedback

Month 3

- Polish UX based on beta feedback
- Add SendGrid relay for users without SMTP setup
- Basic Admin portal: user list, email volume monitor
- Set up Stripe for billing, launch free + paid plans
- Apply for LinkedIn Jobs API partner access

This document is a living roadmap. Revisit and update it at the end of each phase based on user feedback, market signals, and financial performance. The most important step is to **ship Phase 1** — everything else follows from real-world validation.

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